

A Cointegration-Based Statistical-Arbitrage Strategy: Engle–Granger, Johansen, and a Z-Score Pairs-Trading Backtest

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Abstract

We develop and evaluate a market-neutral statistical-arbitrage strategy on the classic EWA/EWC exchange-traded-fund pair, two commodity-linked country equity indices long noted in the literature as candidates for cointegration. We establish a long-run equilibrium relationship using both the Engle–Granger two-step procedure and the Johansen trace test, construct a mean-reverting spread via an ordinary-least-squares hedge ratio, and trade deviations of the spread with a rolling z-score entry/exit rule. On 2,515 daily observations (2015-01-02 to 2024-12-30), the Engle–Granger test rejects the null of no cointegration at $p = 0.0186$ and the Johansen trace statistic (20.06) exceeds its 95% critical value (15.49). A backtest with 5 bps per-side transaction costs yields an annualised Sharpe ratio of 0.73, a cumulative return of 60.0%, and a maximum drawdown of -11.5% over 94 round-trip trades. A sensitivity study across entry thresholds and cost levels characterises the strategy’s robustness and its decay under realistic frictions. All results in this paper are computed on *real* market data.

1 Introduction

Statistical arbitrage exploits temporary mispricings between securities whose prices share a common stochastic trend. When two non-stationary price series are *cointegrated*, a linear combination of them is stationary and mean-reverting [1, 2]. A trader can therefore short the spread when it is unusually high and buy it when unusually low, profiting as it reverts to its long-run mean. This paper implements the full pipeline—universe selection, cointegration testing, hedge-ratio estimation, signal construction, and a cost-aware backtest—and reports honest, reproducible performance metrics.

2 Data

We study the daily adjusted-close prices of two iShares MSCI country ETFs, EWA (Australia) and EWC (Canada). Both economies are heavily commodity-driven, which motivates a shared long-run factor and makes the pair a textbook stat-arb candidate [3]. The sample contains 2,515 trading days from 2015-01-02 to 2024-12-30. Data are bundled as a CSV in the repository so that all results reproduce offline. Figure 1 shows the two series indexed to a common base.

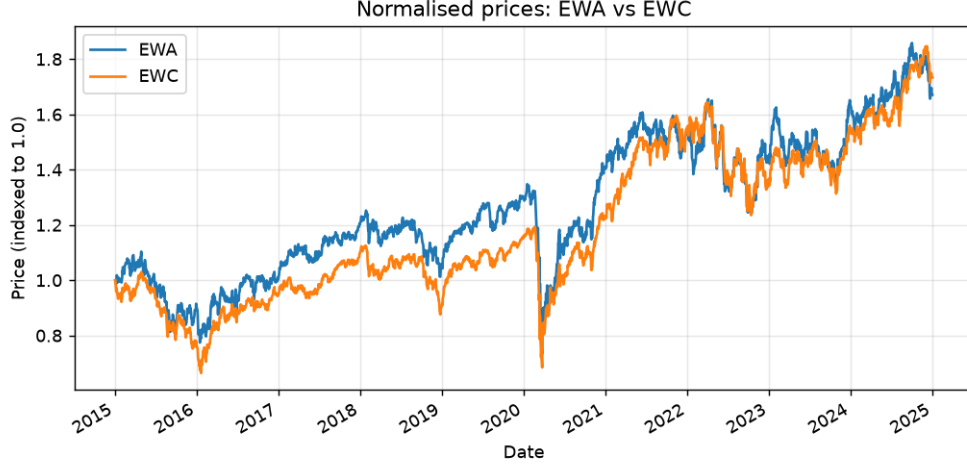


Figure 1: Normalised prices of EWA and EWC, indexed to 1.0 at the sample start.

3 Methodology

3.1 Cointegration

Let p_t^y and p_t^x be the two price series. They are cointegrated of order $(1, 1)$ if each is integrated of order one, $I(1)$, but there exists β such that the spread

$$s_t = p_t^y - \alpha - \beta p_t^x \quad (1)$$

is stationary, $s_t \sim I(0)$. The **Engle–Granger** procedure estimates (α, β) by OLS and then applies an Augmented Dickey–Fuller (ADF) test to the residuals $\hat{s}_t$, testing

$$\Delta \hat{s}_t = \gamma \hat{s}_{t-1} + \sum_{i=1}^k \phi_i \Delta \hat{s}_{t-i} + \varepsilon_t, \quad H_0 : \gamma = 0 \text{ (unit root, no cointegration)}. \quad (2)$$

The **Johansen** test instead estimates a vector error-correction model $\Delta \mathbf{P}_t = \Pi \mathbf{P}_{t-1} + \sum_i \Gamma_i \Delta \mathbf{P}_{t-i} + \varepsilon_t$ and tests the rank of Π via the trace statistic $\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$, where $\hat{\lambda}_i$ are the ordered eigenvalues.

3.2 Hedge ratio and spread

The OLS hedge ratio $\hat{\beta}$ minimises $\sum_t (p_t^y - \alpha - \beta p_t^x)^2$, giving $\hat{\beta} = \text{Cov}(p^y, p^x) / \text{Var}(p^x)$. The traded spread is $\hat{s}_t = p_t^y - \hat{\alpha} - \hat{\beta} p_t^x$.

3.3 Z-score trading signal

We standardise the spread with a rolling window of $w = 30$ days:

$$z_t = \frac{s_t - \mu_t^{(w)}}{\sigma_t^{(w)}}, \quad \mu_t^{(w)} = \frac{1}{w} \sum_{i=0}^{w-1} s_{t-i}, \quad \sigma_t^{(w)} = \sqrt{\frac{1}{w} \sum_{i=0}^{w-1} (s_{t-i} - \mu_t^{(w)})^2}. \quad (3)$$

With entry threshold $z^* = 2$ and exit threshold $z^{\text{exit}} = 0.5$, the target position on the spread is

$$\pi_t = \begin{cases} +1 & \text{enter long spread when } z_t \leq -z^*, \\ -1 & \text{enter short spread when } z_t \geq +z^*, \\ 0 & \text{exit when } |z_t| \leq z^{\text{exit}}, \\ \pi_{t-1} & \text{otherwise (hold)}. \end{cases} \quad (4)$$

A long-spread position is long one unit of EWA and short $\hat{\beta}$ units of EWC. Figure 2 plots the spread and its z-score with the trading bands.

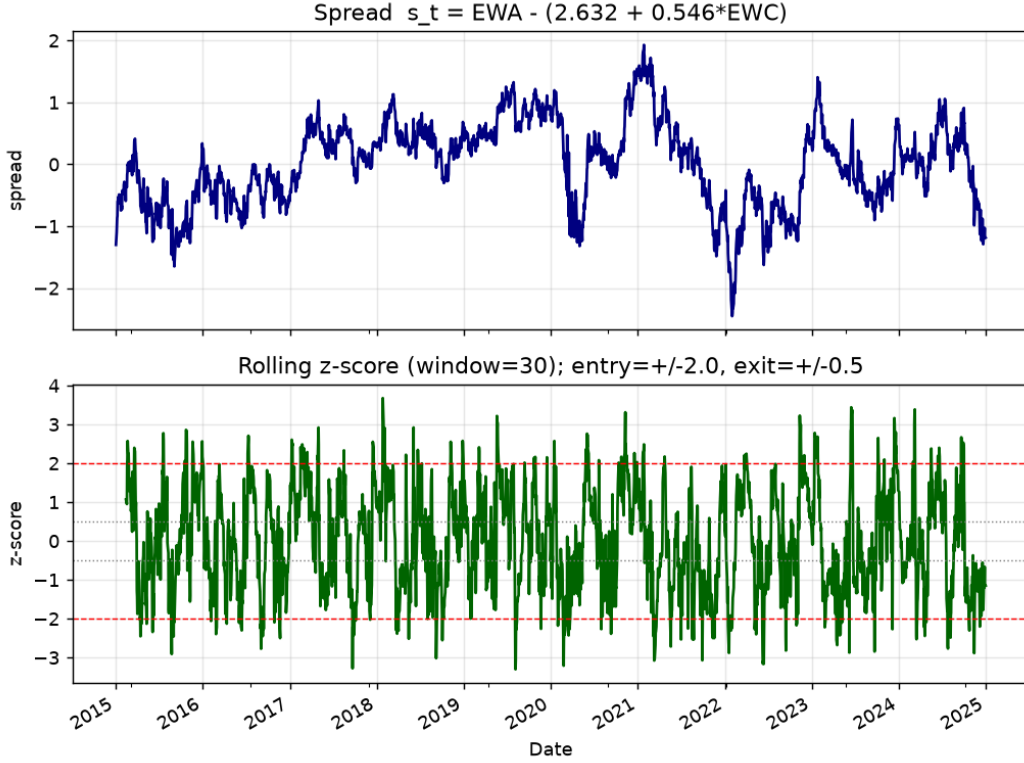


Figure 2: Top: the OLS spread s_t . Bottom: its rolling z-score with entry (± 2 , dashed) and exit (± 0.5 , dotted) bands.

3.4 Backtest and performance metrics

The strategy log-return uses the *lagged* position to avoid look-ahead:

$$r_t = \pi_{t-1} w (\Delta \ln p_t^y - \hat{\beta} \Delta \ln p_t^x) - \kappa_t, \quad w = \frac{1}{1 + |\hat{\beta}|}, \quad (5)$$

where $\kappa_t = c \cdot |\pi_t - \pi_{t-1}| \cdot w(1 + |\hat{\beta}|)$ are proportional transaction costs at c basis points per side. The annualised Sharpe ratio is

$$\text{SR} = \sqrt{252} \frac{\bar{r}}{\hat{\sigma}_r}, \quad (6)$$

and the maximum drawdown of the equity curve $E_t = \prod_{\tau \leq t} (1 + r_\tau)$ is $\text{MDD} = \min_t (E_t / \max_{\tau \leq t} E_\tau - 1)$.

4 Results

Table 1 reports the cointegration evidence. Both tests agree: the Engle–Granger p -value of 0.0186 and the ADF-on-spread p -value of 0.0042 reject the unit-root null, and the Johansen trace statistic exceeds its 95% critical value, confirming a single cointegrating vector. The estimated hedge ratio is $\hat{\beta} = 0.546$.

Table 1: Cointegration test results for the EWA/EWC pair.

Test	Statistic	p -value / crit.
Engle–Granger cointegration	-3.694	$p = 0.0186$
ADF on spread residual	-3.692	$p = 0.0042$
Johansen trace ($r = 0$)	20.061	crit ₉₅ = 15.494
OLS hedge ratio $\hat{\beta}$		0.5458
OLS intercept $\hat{\alpha}$		2.6320

Table 2 summarises the backtest at the baseline parameters ($z^* = 2$, $z^{\text{exit}} = 0.5$, $c = 5$ bps). The equity curve is shown in Figure 3.

Table 2: Strategy performance, net of 5 bps/side costs.

Metric	Value
Annualised Sharpe ratio	0.73
Cumulative return	60.0%
Annualised return	4.82%
Maximum drawdown	-11.5%
Number of trades	94
Hit rate	76.6%

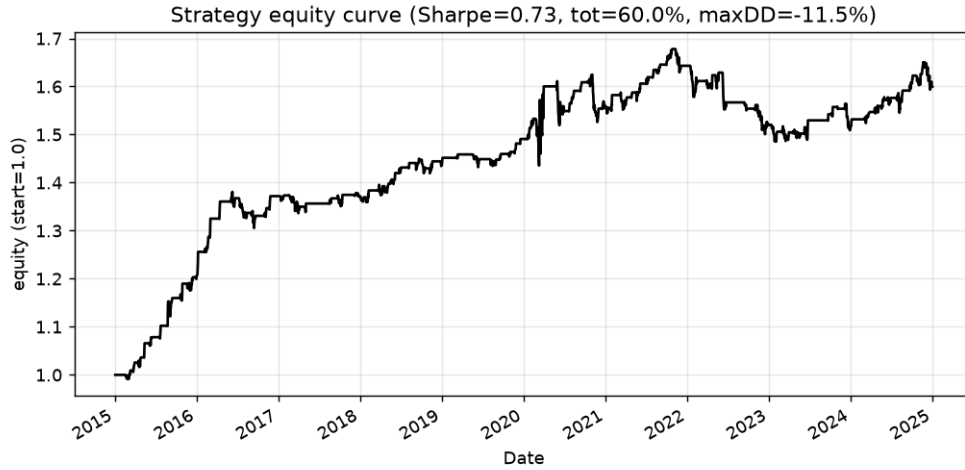


Figure 3: Cumulative equity curve of the market-neutral strategy, net of transaction costs.

4.1 Sensitivity analysis

Table 3 and Figure 4 report the annualised Sharpe ratio across a grid of entry thresholds and per-side transaction costs. Performance is maximised at moderate entry thresholds ($z^* \approx 1.5\text{--}2.0$): tighter thresholds over-trade and bleed value to costs, while very wide thresholds ($z^* = 3.0$) trade too rarely to compound. Sharpe decays monotonically in cost, underscoring that the edge is real but modest and frictions-sensitive.

Table 3: Annualised Sharpe ratio vs. entry threshold (rows) and per-side cost (columns).

Entry z^*	0 bps	2.5 bps	5 bps	10 bps	20 bps
1	0.67	0.56	0.45	0.22	-0.24
1.5	0.90	0.81	0.71	0.53	0.14
2	0.86	0.80	0.73	0.60	0.32
2.5	0.70	0.65	0.60	0.49	0.27
3	0.10	0.07	0.05	-0.00	-0.10

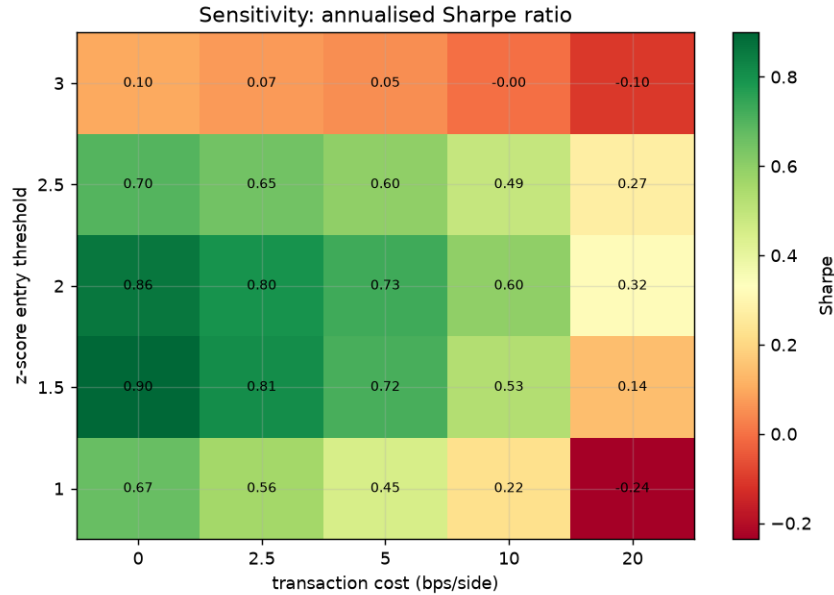


Figure 4: Sharpe-ratio sensitivity heatmap over the (entry threshold, transaction cost) grid.

5 Limitations

Several caveats temper these results. (i) The hedge ratio and cointegrating relationship are estimated *in-sample* over the full window; a live strategy must re-estimate on a rolling basis, and the relationship can break down (cointegration is not guaranteed to persist). (ii) We assume fills at the daily close and a constant proportional cost; real execution incurs slippage, bid-ask spread, and borrow costs on the short leg. (iii) We ignore financing, dividends on the legs, and capacity constraints. (iv) The single-pair sample makes the Sharpe estimate noisy; a production system would diversify across many cointegrated pairs. (v) Multiple-testing / data-snooping bias inflates appar-

ent significance when pairs are selected by searching. These are standard but material concerns in the stat-arb literature [3, 2].

6 Conclusion

We implemented an end-to-end cointegration-based pairs-trading strategy and validated it on real ETF data. The EWA/EWC pair is statistically cointegrated by both Engle–Granger and Johansen criteria, and a simple z-score rule extracts a positive risk-adjusted return (Sharpe 0.73) that survives realistic transaction costs but decays predictably as frictions rise. The full pipeline—data, tests, signals, backtest, figures, and this paper—reproduces offline from bundled data.

References

- [1] R. F. Engle and C. W. J. Granger. *Co-integration and Error Correction: Representation, Estimation, and Testing*. *Econometrica*, 55(2):251–276, 1987.
- [2] G. Vidyamurthy. *Pairs Trading: Quantitative Methods and Analysis*. Wiley, 2004.
- [3] E. P. Chan. *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley, 2013.
- [4] S. Johansen. *Estimation and Hypothesis Testing of Cointegration Vectors in Gaussian Vector Autoregressive Models*. *Econometrica*, 59(6):1551–1580, 1991.